

WEEK-12 –Creating IAM USER with permission Access it with GUI and CLI

Part A: GUI Access (Management Console)

Objective: Create an IAM user with restricted permissions and verify access via the AWS web portal.

1. Administrative Setup (Root User)

- **Step 1:** Log in to the AWS Management Console as the **Root User**.
- **Step 2:** Search for **IAM** under the "Security, Identity, & Compliance" category.
- **Step 3:** Navigate to Users and click Create user.
- **Step 4:** User Details: Enter a name (e.g., S3_Specialist). Check the box for "Provide user access to the AWS Management Console." – custom password.
- **Step 5:** Set Permissions: Select "Attach policies directly." Search for and select AmazonS3FullAccess.
- **Step 6:** Review & Create: Complete the wizard. On the final page, click Download .csv file.

Note: This file contains your Console Password and the unique Sign-in URL.

- **Step 7:** Copy your 12-digit Account ID (found in the top-right dropdown or the .csv) and Sign Out.

2. Verification (IAM User Login)

- **Step 8:** Use the Sign-in URL from the .csv (or go to the AWS sign-in page and select **IAM User**).
- **Step 9:** Enter the **Account ID**, then the **IAM Username** and **Password** from the .csv.
- **Step 10: The Logic Test:**
 - **Test 1 (Failure):** Navigate to **EC2**. Try to view instances. You will see "API Error" or "Access Denied." This proves the user is restricted.
 - **Test 2 (Success):** Navigate to **S3**. Create a new bucket. It will work perfectly because of the attached policy.

Part B: CLI Access (Programmatic Access)

Objective: Access AWS resources from a standalone system (CMD/Terminal) using security keys.

1. Credential Generation

- **Step 1:** While logged in as an Admin/Root, go to **IAM > Users** and click on your user.
- **Step 2:** Go to the **Security credentials** tab. Scroll to **Access keys** and click **Create access key**.
- **Step 3:** Select **Command Line Interface (CLI)**. Acknowledge the warning and click **Next**.
- **Step 4: Download the .csv file.** > **Critical:** This contains the **Access Key ID** and **Secret Access Key**. If lost, you must delete and recreate them.

2. Standalone System Configuration

- **Step 5:** Install the **AWS CLI Tool** from google on your computer.
- **Step 6:** Open your Command Prompt (CMD) or Terminal.
- **Step 7:** Type `aws configure` and press Enter.
- **Step 8:** Provide the following from your .csv:
 - **Access Key ID**
 - **Secret Access Key**
 - **Default region:** (e.g., `ap-south-1`)
 - **Default output:** `json`

3. Verification via Commands

- **Step 9:** Run `aws s3 ls`. This should list your buckets.
- **Step 10:** Run `aws s3 mb s3://your-unique-bucket-name`. (Success)
- **Step 11:** Run `aws iam list-users`.
 - **Result:** You will get an **(AccessDenied)** error.

This confirms that the CLI is respecting the same IAM policy as the GUI.